

1. Write a Java Program to display a “Hello World”

Main:

```
public class JavaApplication {  
    public static void main(String[] args) {  
        HelloWorld hello = new HelloWorld();  
        hello.display();  
    }  
}
```

Class:

```
public class HelloWorld {  
    public void display(){  
        System.out.println("Hello World");  
    }  
}
```

Output:

it - JavaApplication35 (run)

```
run:  
Hello World  
BUILD SUCCESSFUL (total time: 0 seconds)
```

2. Write a Java Program to find the sum of two integer numbers with user input and display the result

Main:

```
public class JavaApplication {  
    public static void main(String[] args) {  
        SumOf sumof = new SumOf();  
        sumof.AddTwo();  
    }  
}
```

Class:

```
import java.util.Scanner;  
  
public class SumOf {  
    int num1, num2;  
    public void AddTwo(){  
        Scanner scanner = new Scanner(System.in);  
  
        System.out.println("enter first number: ");  
        num1 = scanner.nextInt();
```

```

        System.out.println("enter second number: ");
        num2 = scanner.nextInt();

        System.out.println("Sumf of "+num1+" and "+num2+" is "+ (num1+num2));
    }
}

```

Output:

```

t - JavaApplication35 (run)

run:
enter first number:
23
enter second number:
66
Sumf of 23 and 66 is 89
BUILD SUCCESSFUL (total time: 3 seconds)

```

3. Write a java program to find the simple interest using user input

Main:

```

public class JavaApplication {
    public static void main(String[] args) {
        SimpleInterest simple = new SimpleInterest();
        simple.Interest();
    }
}

```

Class:

```

import java.util.Scanner;

public class SimpleInterest {
    int time,principle;
    float rate;
    public void Interest(){
        Scanner scanner = new Scanner(System.in);
        System.out.println("enter principle number: ");
        principle = scanner.nextInt();
        System.out.println("enter time number: ");
    }
}

```

```

        time = scanner.nextInt();
        System.out.println("enter rate number: ");
        rate = scanner.nextFloat();
        System.out.println("Simple interest = "+(principle*time*rate)/100);
    }
}

```

Output:

```

ut - JavaApplication35 (run)

enter principle number:
50000
enter time number:
3
enter rate number:
6
Simple interest = 9000.0
BUILD SUCCESSFUL (total time: 7 seconds)

```

4. Write a java program to find the Compound interest

Main:

```

public class JavaApplication {
    public static void main(String[] args) {
        CompoundInterest compound = new CompoundInterest();
        compound.Interest();
    }
}

```

Class:

```

import java.util.Scanner;

public class CompoundInterest {
    int time, principle;
    double rate;

    public void Interest() {
        Scanner scanner = new Scanner(System.in);

        System.out.print("enter principle number: ");
    }
}

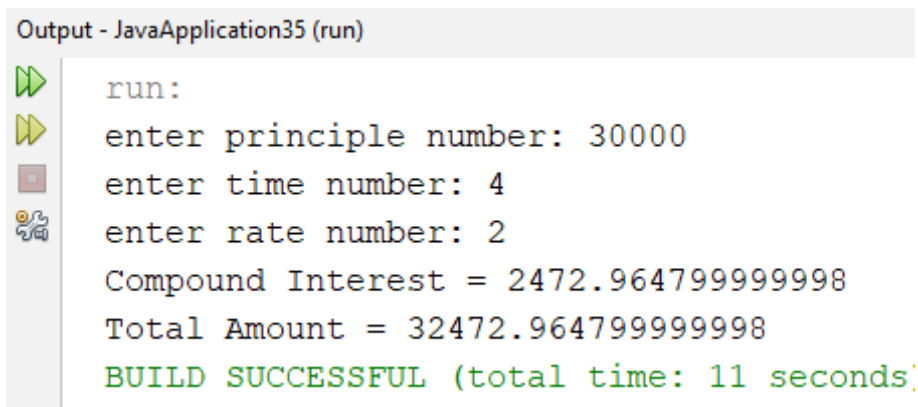
```

```

        principle = scanner.nextInt();
        System.out.print("enter time number: ");
        time = scanner.nextInt();
        System.out.print("enter rate number: ");
        rate = scanner.nextDouble();
        double amount = principle * Math.pow((1 + rate / 100), time);
        double ci = amount - principle;
        System.out.println("Compound Interest = " + ci);
        System.out.println("Total Amount = " + amount);
    }
}

```

Output:



```

Output - JavaApplication35 (run)

run:
enter principle number: 30000
enter time number: 4
enter rate number: 2
Compound Interest = 2472.9647999999998
Total Amount = 32472.964799999998
BUILD SUCCESSFUL (total time: 11 seconds)

```

5. Write a java program to print your name,gender,DOB with user input

Main:

```

public class JavaApplication {
    public static void main(String[] args) {
        UserDetails userdetail = new UserDetails();
        userdetail.printDetails();
    }
}

```

Class:

```

import java.util.Scanner;
public class UserDetails {
    String name, gender, dob;
}

```

```

public void printDetails() {
    Scanner scanner = new Scanner(System.in);

    System.out.print("Enter your Name: ");
    name = scanner.nextLine();

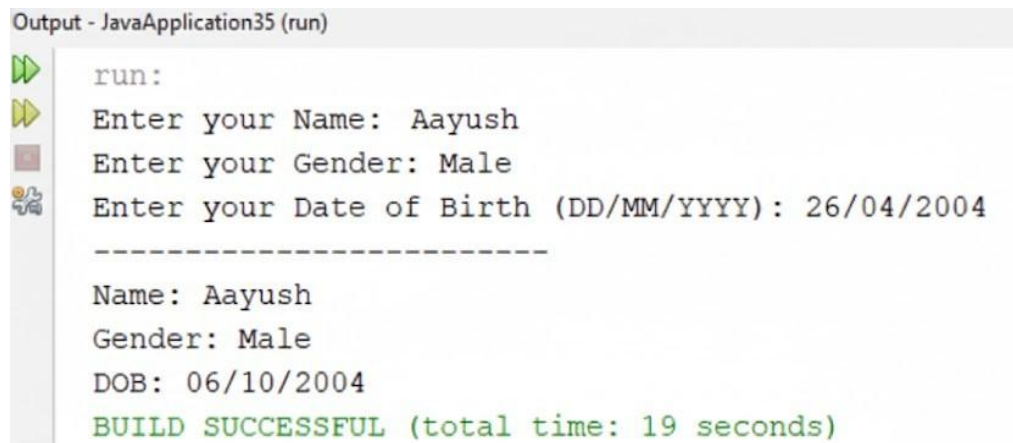
    System.out.print("Enter your Gender: ");
    gender = scanner.nextLine();

    System.out.print("Enter your Date of Birth (DD/MM/YYYY): ");
    dob = scanner.nextLine();

    System.out.println("-----");
    System.out.println("Name: " + name);
    System.out.println("Gender: " + gender);
    System.out.println("DOB: " + dob);
}
}

```

Output:



```

Output - JavaApplication35 (run)

run:
Enter your Name: Aayush
Enter your Gender: Male
Enter your Date of Birth (DD/MM/YYYY): 26/04/2004
-----
Name: Aayush
Gender: Male
DOB: 06/10/2004
BUILD SUCCESSFUL (total time: 19 seconds)

```

6. Write a java program to find the roots of Quadratic equation. $X = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. where a,b and c should be a user input

Main:

```

public class JavaApplication {
    public static void main(String[] args) {
        QuadraticSolver quad = new QuadraticSolver();
        quad.findRoots();
    }
}

```

Class:

```

import java.util.Scanner;

public class QuadraticSolver {
    double a, b, c;

    public void findRoots() {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter value for a: ");
        a = scanner.nextDouble();
        System.out.print("Enter value for b: ");
        b = scanner.nextDouble();
        System.out.print("Enter value for c: ");
        c = scanner.nextDouble();
        double discriminant = (b * b) - (4 * a * c);
        if (discriminant > 0) {
            double root1 = (-b + Math.sqrt(discriminant)) / (2 * a);
            double root2 = (-b - Math.sqrt(discriminant)) / (2 * a);

            System.out.println("Roots are real and different.");
            System.out.println("Root 1 = " + root1);
            System.out.println("Root 2 = " + root2);

        } else if (discriminant == 0) {
            double root = -b / (2 * a);

            System.out.println("Roots are real and equal.");
            System.out.println("Root = " + root);

        } else {
            double realPart = -b / (2 * a);
            double imaginaryPart = Math.sqrt(-discriminant) / (2 * a);

            System.out.println("Roots are complex (imaginary).");
            System.out.printf("Root 1 = %.2f + %.2fi %n", realPart, imaginaryPart);
            System.out.printf("Root 2 = %.2f - %.2fi %n", realPart, imaginaryPart);
        }
    }
}

```

Output:

```
ut - JavaApplication35 (run)

run:
Enter value for a: 3
Enter value for b: 6
Enter value for c: 2
Roots are real and different.
Root 1 = -0.42264973081037427
Root 2 = -1.5773502691896255
BUILD SUCCESSFUL (total time: 10 seconds)
```

7. Write a program to find the greatest number between two numbers (a,b) where these two numbers should be the user input

Main:

```
public class JavaApplication {
    public static void main(String[] args) {
        NumberCompare numcom = new NumberCompare();
        numcom.findGreatest();
    }
}
```

Class:

```
import java.util.Scanner;
```

```
public class NumberCompare {
    int a, b;

    public void findGreatest() {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter the first number (a): ");
        a = scanner.nextInt();

        System.out.print("Enter the second number (b): ");
        b = scanner.nextInt();

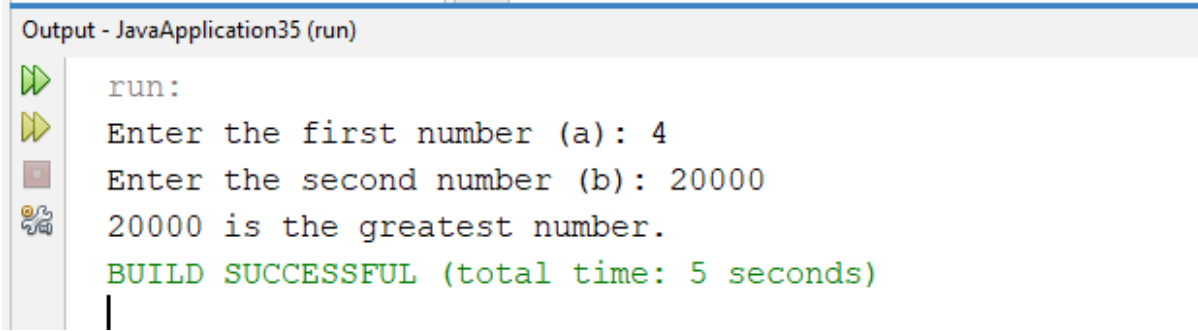
        if (a > b) {
            System.out.println(a + " is the greatest number.");
        } else if (b > a) {
            System.out.println(b + " is the greatest number.");
        } else {
            System.out.println("Both numbers are equal.");
        }
    }
}
```

```

    }
}
}

```

Output:



```

Output - JavaApplication35 (run)
run:
Enter the first number (a): 4
Enter the second number (b): 20000
20000 is the greatest number.
BUILD SUCCESSFUL (total time: 5 seconds)
|

```

8. Write a program to find the lowest number between three numbers(a,b,c) where these numbers should be the use input

Main:

```

public class JavaApplication {
    public static void main(String[] args) {
        NumberFinder numfind = new NumberFinder();
        numfind.findLowest();
    }
}

```

Class:

```
import java.util.Scanner;
```

```
public class NumberFinder {
    int a, b, c;

```

```
    public void findLowest() {
        Scanner scanner = new Scanner(System.in);

```

```
        System.out.print("Enter the first number (a): ");
        a = scanner.nextInt();

```

```
        System.out.print("Enter the second number (b): ");
        b = scanner.nextInt();

```

```
        System.out.print("Enter the third number (c): ");
        c = scanner.nextInt();

```

```
        int lowest = a;

```

```

        if (b < lowest) {
            lowest = b;
        }

        if (c < lowest) {
            lowest = c;
        }
        System.out.println("The lowest number is: " + lowest);
    }
}

```

Output:

```

out - JavaApplication35 (run)

run:
Enter the first number (a): 34
Enter the second number (b): 54
Enter the third number (c): 22
The lowest number is: 22
BUILD SUCCESSFUL (total time: 5 seconds)

```

9. Write a program to generate the simple result where the user must enter the marks of five subjects then check whether the given entered marks are passed or not. If marks entered in all subjects is greater than pass marks then calculate the percentage and division.(assume P.M=35,F.M=100)

Main:

```

public class JavaApplication {
    public static void main(String[] args) { ResultGenerator
        result = new ResultGenerator(); result.generateResult();}
}

```

Class:

```

import java.util.Scanner;

public class ResultGenerator {
    double sub1, sub2, sub3, sub4, sub5;
    double passMark = 35;
    double fullMarkPerSubject = 100;
    public void generateResult() {
        Scanner scanner = new Scanner(System.in);

        // 1. Input Marks
        System.out.println("Enter marks for 5 subjects (out of 100):");
        System.out.print("Subject 1: ");
        sub1 = scanner.nextDouble();
        System.out.print("Subject 2: ");
        sub2 = scanner.nextDouble();
        System.out.print("Subject 3: ");

```

```

sub3 = scanner.nextDouble();
System.out.print("Subject 4: ");
sub4 = scanner.nextDouble();
System.out.print("Subject 5: ");
sub5 = scanner.nextDouble();

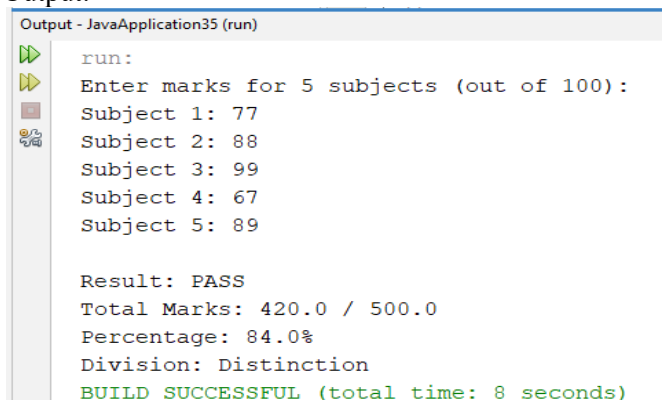
if (sub1 < passMark || sub2 < passMark || sub3 < passMark || sub4 < passMark || sub5 < passMark) {
    System.out.println("\nResult: FAIL");
    System.out.println("Reason: One or more subjects are below the pass mark of " + passMark);
} else {
    double totalObtained = sub1 + sub2 + sub3 + sub4 + sub5;
    double totalMaxMarks = fullMarkPerSubject * 5;
    double percentage = (totalObtained / totalMaxMarks) * 100;

    System.out.println("\nResult: PASS");
    System.out.println("Total Marks: " + totalObtained + " / " + totalMaxMarks);
    System.out.println("Percentage: " + percentage + "%");

    String division;
    if (percentage >= 80) {
        division = "Distinction";
    } else if (percentage >= 60) {
        division = "First Division";
    } else if (percentage >= 45) {
        division = "Second Division";
    } else {
        division = "Third Division";
    }
    System.out.println("Division: " + division);
}
}
}

```

Output:



```

Output - JavaApplication35 (run)
run:
Enter marks for 5 subjects (out of 100):
Subject 1: 77
Subject 2: 88
Subject 3: 99
Subject 4: 67
Subject 5: 89

Result: PASS
Total Marks: 420.0 / 500.0
Percentage: 84.0%
Division: Distinction
BUILD SUCCESSFUL (total time: 8 seconds)

```

10. Write a Java program to find the number of days in a month

Main:

```
public class JavaApplication {
```

```

        public static void main(String[] args) {
            MonthDays month = new MonthDays();
            month.findDays();}}

```

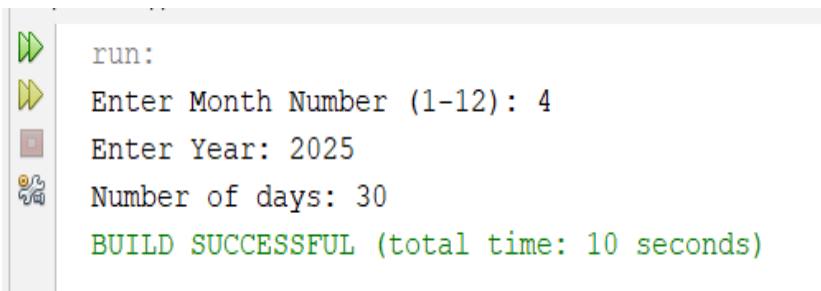
Class:

```

import java.util.Scanner;
public class MonthDays {
    int month, year;
    public void findDays() {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter Month Number (1-12): ");
        month = scanner.nextInt();
        System.out.print("Enter Year: ");
        year = scanner.nextInt();
        switch (month) {
            // Months with 31 Days
            case 1: // January
            case 3: // March
            case 5: // May
            case 7: // July
            case 8: // August
            case 10: // October
            case 12: // December
                System.out.println("Number of days: 31");
                break;
            // Months with 30 Days
            case 4: // April
            case 6: // June
            case 9: // September
            case 11: // November
                System.out.println("Number of days: 30");
                break;
            // February
            case 2:
                // Check for Leap Year
                if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
                    System.out.println("Number of days: 29 (Leap Year)");
                } else {
                    System.out.println("Number of days: 28");
                }
                break;
            default:
                System.out.println("Invalid month! Please enter 1-12.");
        }
    }
}

```

Output:



```
run:
Enter Month Number (1-12): 4
Enter Year: 2025
Number of days: 30
BUILD SUCCESSFUL (total time: 10 seconds)
```

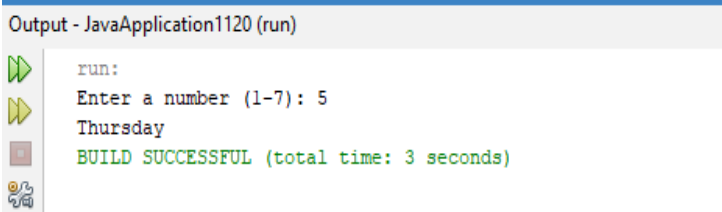
11) Write a program to find the day when we enter the number using a switch case. Where number should be entered by user and if you press

1 then display Sunday

2 then display Monday

3 then display Tuesday and so on...

```
import java.util.Scanner;
class DayFinder {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number (1-7): ");
        int day = sc.nextInt();
        switch(day) {
            case 1: System.out.println("Sunday"); break;
            case 2: System.out.println("Monday"); break;
            case 3: System.out.println("Tuesday"); break;
            case 4: System.out.println("Wednesday"); break;
            case 5: System.out.println("Thursday"); break;
            case 6: System.out.println("Friday"); break;
            case 7: System.out.println("Saturday"); break;
            default: System.out.println("Invalid input"); } } }
```



```
run:
Enter a number (1-7): 5
Thursday
BUILD SUCCESSFUL (total time: 3 seconds)
```

12) Write a program to find sum, subtract, multiply and division using a switch case.

Where two numbers must be user input and user also must enter the operator.if your press

+ then two numbers must be added and display the result.

- then two numbers must be subtracted and display the result.

*** then two numbers must be multiplied and display the result.**

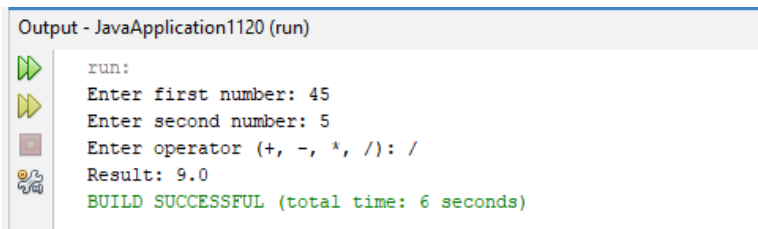
/ then there should be division of numbers

```
import java.util.Scanner;
class CalculatorSwitch {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first number: ");
        double a = sc.nextDouble();
        System.out.print("Enter second number: ");
```

```

double b = sc.nextDouble();
System.out.print("Enter operator (+, -, *, /): ");
char op = sc.next().charAt(0);
switch(op) {
    case '+': System.out.println("Result = " + (a + b)); break;
    case '-': System.out.println("Result = " + (a - b)); break;
    case '*': System.out.println("Result = " + (a * b)); break;
    case '/':
        if(b != 0)
            System.out.println("Result = " + (a / b));
        else
            System.out.println("Division by zero error");
        break;
    default: System.out.println("Invalid operator"); } }

```



```

Output - JavaApplication1120 (run)
run:
Enter first number: 45
Enter second number: 5
Enter operator (+, -, *, /): /
Result: 9.0
BUILD SUCCESSFUL (total time: 6 seconds)

```

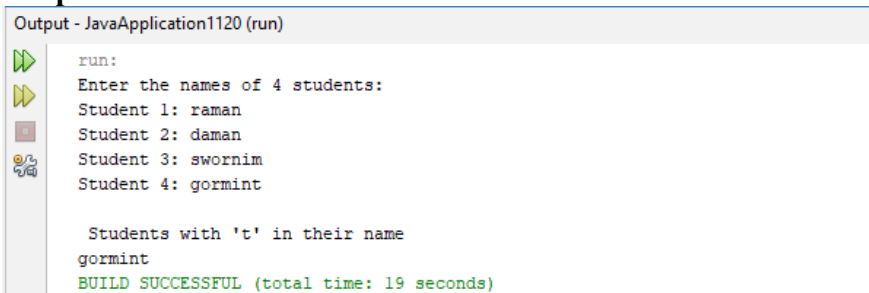
13) Define String array of size 4 and store name of 4 students .The display the names of students whose name has character t.

```

import java.util.Scanner;
class NamesWithT {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String[] names = new String[4];
        for(int i=0; i<4; i++) {
            System.out.print("Enter name of student " + (i+1) + ": ");
            names[i] = sc.nextLine();}
        System.out.println("Names containing 't':");
        for(String name : names) {
            if(name.toLowerCase().contains("t")) {
                System.out.println(name);} } }

```

Output:



```

Output - JavaApplication1120 (run)
run:
Enter the names of 4 students:
Student 1: raman
Student 2: daman
Student 3: swornim
Student 4: gormint

Students with 't' in their name
gormint
BUILD SUCCESSFUL (total time: 19 seconds)

```

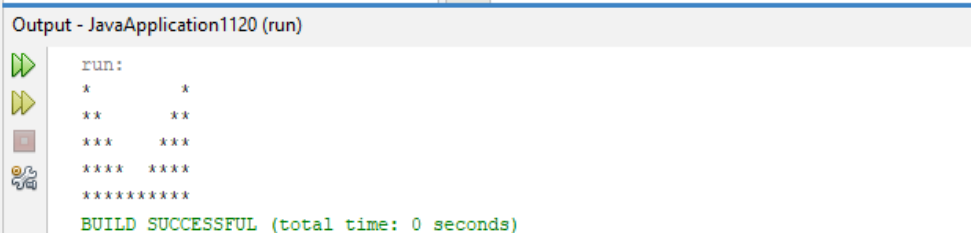
14) Program to print the following pattern

```
*      *
**     **
***    ***
****   ****
*****
```

Main

```
class StarPatternDynamic {
    public static void main(String[] args) {
        int n = 5; // number of rows
        for (int i = 1; i <= n; i++) {
            // Print left stars
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            int spaces;
            if (i == n) {
                spaces = 0; // last row has no space
            } else {
                spaces = 2 * (n - i); // middle spaces
            }
            for (int j = 1; j <= spaces; j++) {
                System.out.print(" ");
            }
            // Print right stars
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println(); // next row
        }
    }
}
```

Output:

A screenshot of a Java IDE's output window. The title bar reads "Output - JavaApplication1120 (run)". On the left, there are icons for running, stepping through, and debugging. The output text shows the pattern of stars and spaces from the previous program, followed by a green status message: "BUILD SUCCESSFUL (total time: 0 seconds)".

```
run:
*      *
**     **
***    ***
****   ****
*****
BUILD SUCCESSFUL (total time: 0 seconds)
```

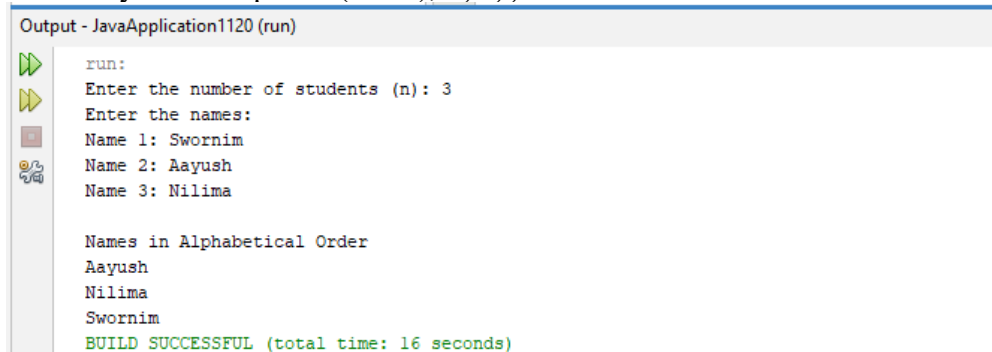
15) Create an String array of size n then enter the element in array and print the name in alphabetical order

```
import java.util.Scanner;
import java.util.Arrays;
class SortStrings {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = sc.nextInt();
        sc.nextLine(); // consume newline
        String[] arr = new String[n];
        for(int i=0; i<n; i++) {
            System.out.print("Enter name " + (i+1) + ": ");
            arr[i] = sc.nextLine();
        }
    }
}
```

```

Arrays.sort(arr);
System.out.println("Names in alphabetical order:");
for(String name : arr) {
    System.out.println(name); } }

```



```

Output - JavaApplication1120 (run)

run:
Enter the number of students (n): 3
Enter the names:
Name 1: Swornim
Name 2: Aayush
Name 3: Nilima

Names in Alphabetical Order
Aayush
Nilima
Swornim
BUILD SUCCESSFUL (total time: 16 seconds)

```

16) Write a program to sort an integer array elements

Main:

```

public class JavaApplication1120 {
    public static void main(String[] args) {
        ArraySorter sort = new ArraySorter();
        sort.sortNumbers();} }

```

Class:

```

import java.util.Scanner;
public class ArraySorter {
    public void sorting(int[] numbers){
        for(int i = 0; i<numbers.length; i++){
            for(int j = i+1; j<numbers.length; j++){
                if(numbers[i]>numbers[j]){
                    int temp = numbers[i];
                    numbers[i] = numbers[j];
                    numbers[j] = temp; } } } }
    public void sortNumbers() {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = scanner.nextInt();
        int[] numbers = new int[n];
        System.out.println("Enter " + n + " integers:");
        for (int i = 0; i < n; i++) {
            System.out.print("Element " + (i + 1) + ": ");
            numbers[i] = scanner.nextInt();
        }
        sorting(numbers);
        System.out.println("\nSorted Array:");
        for (int num : numbers) {
            System.out.print(num + " ");
        }
        System.out.println();
    }
}

```

```
Output - JavaApplication1120 (run)

run:
Enter the number of elements: 7
Enter 7 integers:
Element 1: 34
Element 2: 23
Element 3: 54
Element 4: 65
Element 5: 12
Element 6: 90
Element 7: 13

Sorted Array:
12 13 23 34 54 65 90
BUILD SUCCESSFUL (total time: 14 seconds)
```

17) Create an array of size n then enter the element in array and find the sum of array elements

```
import java.util.Scanner;
class ArraySum {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        int sum = 0;
        System.out.print("Enter"+i+"numbers": ");
        for(int i=0; i<n; i++) {
            arr[i] = sc.nextInt();
            sum += arr[i]; }
        System.out.println("Sum = " + sum);}}
```

```
Output - JavaApplication1120 (run)

run:
Enter the number of elements (n): 5
Enter 5 numbers:
54
6656
43
43
22
Sum = 6818
BUILD SUCCESSFUL (total time: 12 seconds)
```

18) Create an array of size n then enter the element in the array and find min value and max value of an array.

```
import java.util.Scanner;
class MinMaxArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = sc.nextInt();

        int[] arr = new int[n];
        for(int i=0; i<n; i++) {
            System.out.print("Enter element " + (i+1) + ": ");
            arr[i] = sc.nextInt();}
        int min = arr[0], max = arr[0];
```

```

        for(int i=1; i<n; i++) {
            if(arr[i] < min) min = arr[i];
            if(arr[i] > max) max = arr[i];}
        System.out.println("Min = " + min);
        System.out.println("Max = " + max); }}
Enter size of array: 4
Enter element 1: 100
Enter element 2: 2
Enter element 3: 34
Enter element 4: 45
Min = 2
Max = 100
-----

```

19) Create an 2D array of size 2 x 2 then perform the matrix sum

Main:

```

public class JavaApplication1120 {
    public static void main(String[] args) {
        MatrixSum calc = new MatrixSum();
        calc.addMatrices(); }}

```

Class:

```

import java.util.Scanner;

```

```

public class MatrixSum {

    public void addMatrices() {
        Scanner scanner = new Scanner(System.in);
        int[][] matrix1 = new int[2][2];
        int[][] matrix2 = new int[2][2];
        int[][] sum = new int[2][2];

        System.out.println("Enter elements for First Matrix (2x2):");
        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                System.out.print("Enter element [" + i + "][" + j + "]: ");
                matrix1[i][j] = scanner.nextInt();} }

        System.out.println("\nEnter elements for Second Matrix (2x2):");
        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                System.out.print("Enter element [" + i + "][" + j + "]: ");
                matrix2[i][j] = scanner.nextInt();
            }
        }

        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                sum[i][j] = matrix1[i][j] + matrix2[i][j];} }
        System.out.println("\nSum of Matrices");
        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                System.out.print(sum[i][j] + "\t");
            }
            System.out.println();}}
    }
}

```

```
Output - JavaApplication1120 (run) x
run:
Enter elements for First Matrix (2x2):
Enter element [0][0]: 3
Enter element [0][1]: 4
Enter element [1][0]: 6
Enter element [1][1]: 2

Enter elements for Second Matrix (2x2):
Enter element [0][0]: 21
Enter element [0][1]: 67
Enter element [1][0]: 4
Enter element [1][1]: 88

Sum of Matrices
24      71
10      90

BUILD SUCCESSFUL (total time: 17 seconds)
```

20) Create an 2D array of size 3 x 3 then find the determinant

Main:

```
public class JavaApplication1120 {
    public static void main(String[] args) {
        DeterminantFinder det = new DeterminantFinder();
        det.findDeterminant();
    }
}
```

Class:

```
import java.util.Scanner;
public class DeterminantFinder {
    public void findDeterminant() {
        Scanner scanner = new Scanner(System.in);
        int[][] matrix = new int[3][3];
        System.out.println("Enter elements for the 3x3 Matrix:");
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print("Enter element [" + i + "][" + j + "]: ");
                matrix[i][j] = scanner.nextInt();
            }
        }
        System.out.println("\nYour Matrix:");
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(matrix[i][j] + "\t");
            }
            System.out.println();
        }
        int x = (matrix[1][1] * matrix[2][2]) - (matrix[1][2] * matrix[2][1]);
        int y = (matrix[1][0] * matrix[2][2]) - (matrix[1][2] * matrix[2][0]);
        int z = (matrix[1][0] * matrix[2][1]) - (matrix[1][1] * matrix[2][0]);
        int determinant = (matrix[0][0] * x) - (matrix[0][1] * y) + (matrix[0][2] * z);
        System.out.println("\nDeterminant = " + determinant);
    }
}
```

```
Output - JavaApplication1120 (run) x
run:
Enter elements for the 3x3 Matrix:
Enter element [0][0]: 2
Enter element [0][1]: 4
Enter element [0][2]: 5
Enter element [1][0]: 6
Enter element [1][1]: 7
Enter element [1][2]: 8
Enter element [2][0]: 2
Enter element [2][1]: 3
Enter element [2][2]: 4

Your Matrix:
2      4      5
6      7      8
2      3      4

Determinant = -4
BUILD SUCCESSFUL (total time: 10 seconds)
```

Q.21 Create a 2D array of size 3 x 3 and display the upper triangle.

UpperTriangle.java

```
public class UpperTriangle{
    public static void main(String[] args){
        int[][] a={{1,2,3},{4,5,6},{7,8,9}};
        System.out.println("Upper Triangle Matrix:");
        for(int i=0;i<3;i++){
            for(int j=0;j<3;j++) if(j>=i) System.out.print(a[i][j]+" "); else System.out.print("\t");
            System.out.println();
        }
    }
}
```

Output:

```
--- exec:3.5.1:exec (default-cli) @ AayushAcharyaJava ---
Upper Triangle Matrix:
1      2      3
      5      6
          9
-----
BUILD SUCCESS
```

Q.22 Create a Rectangle class with Set() and Calculate() methods.

RectangleDemo.java

```
import java.util.Scanner;
class Rectangle{
    int length,breadth,area;
    void Set(){
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter length: ");
        length=sc.nextInt();
        System.out.print("Enter breadth: ");
        breadth=sc.nextInt();
    }
    void Calculate(){
        area=length*breadth;
        System.out.println("Area of Rectangle = "+area);
    }
}
public class RectangleDemo{
    public static void main(String[] args){
        Rectangle r=new Rectangle();
        r.Set();r.Calculate();
    }
}
```

Output:

```

--- exec:3.5.1:exec (default-cli) @ AayushAcharyaJava
Enter length: 10
Enter breadth: 20
Area of Rectangle = 200
-----
BUILD SUCCESS

```

Q.23 Create a Quadratic class to input coefficients, calculate roots and display them.

QuadraticDemo.java

```

import java.util.Scanner;
class Quadratic{
    double a,b,c,x1,x2;
    void Input(){
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter value of a: ");a=sc.nextDouble();
        System.out.print("Enter value of b: ");b=sc.nextDouble();
        System.out.print("Enter value of c: ");c=sc.nextDouble(); }
    void Calculate(){
        double d=b*b-4*a*c;
        x1=(-b+Math.sqrt(d))/(2*a);x2=(-b-Math.sqrt(d))/(2*a);}
    void Display(){
        System.out.println("First Root = "+x1);
        System.out.println("Second Root = "+x2); }
    public class QuadraticDemo{
        public static void main(String[] args){
            Quadratic q=new Quadratic();
            q.Input();q.Calculate();q.Display();
        }
    }
}

```

Output:

```

--- exec:3.5.1:exec (default-cli) @ AayushAcharyaJava ---
Enter value of a: Enter value of b: Enter value of c: First Root = 2.0
Second Root = 1.0
-----
BUILD SUCCESS

```

Q.24 Create two Rectangle objects and display the larger area.

RectangleCompare.java

```

class Rectangle{
    int length,breadth,area;
    void computeArea(){area=length*breadth;}
    void displayArea(){System.out.println("Area = "+area);}
}
public class RectangleCompare{
    public static void main(String[] args){
        Rectangle r1=new Rectangle(),r2=new Rectangle();
        r1.length=17;r1.breadth=6;r1.computeArea();
        r2.length=10;r2.breadth=4;r2.computeArea();
        System.out.println("Area of First Rectangle:");r1.displayArea();
        System.out.println("Area of Second Rectangle:");r2.displayArea();
        System.out.println("Larger Rectangle Area = "+Math.max(r1.area,r2.area));
    }
}

```

```
}
```

Output:

```
--- exec:3.5.1:exec (default-cli) @ AayushAcharyaJava ---
```

```
Area of First Rectangle:
Area = 102
Area of Second Rectangle:
Area = 40
Larger Rectangle Area = 102
```

```
-----
BUILD SUCCESS
```

Q.25 Create a Factorial class with int FindFact(int n).

FactorialDemo.java

```
class Factorial{
    int result=1;
    int FindFact(int n){
        for(int i=1;i<=n;i++) result*=i;
        return result;
    }
}
public class FactorialDemo{
    public static void main(String[] args){
        Factorial f=new Factorial();
        System.out.println("Factorial = "+f.FindFact(6));
    }
}
```

Output:

```
--- exec:3.5.1:exec (default-cli) @ AayushAcharyaJava ---
```

```
Factorial = 720
```

```
-----
BUILD SUCCESS
```

Q.26 Create an ArraySum class with int Sum(int[] a).

ArraySumDemo.java

```
class ArraySum{
    int sum=0;
    int Sum(int[] a){
        for(int x:a) sum+=x;
        return sum;}}
public class ArraySumDemo{
    public static void main(String[] args){
        int[] a={1,2,3,4,5,6,7,8,9,10};
        ArraySum obj=new ArraySum();
        System.out.println("Sum = "+obj.Sum(a));
    }
}
```

Output:

```
--- exec:3.5.1:exec (default-cli) @ AayushAcharyaJava ---
```

```
Sum = 55
```

```
-----
BUILD SUCCESS
```

Q.27 Create a Quadratic class where Calculate() returns two roots to main.

QuadraticReturn.java

```
import java.util.Scanner;
class Quadratic{
    double a,b,c,x1,x2;
    void Input(){
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter value of a: ");a=sc.nextDouble();
        System.out.print("Enter value of b: ");b=sc.nextDouble();
        System.out.print("Enter value of c: ");c=sc.nextDouble();
    }
    double[] Calculate(){
        double d=b*b-4*a*c;
        x1=(-b+Math.sqrt(d))/(2*a);x2=(-b-Math.sqrt(d))/(2*a);
        return new double[]{x1,x2};    }
    public class QuadraticReturn{
        public static void main(String[] args){
            Quadratic q=new Quadratic();
            q.Input();
            double[] r=q.Calculate();
            System.out.println("First Root = "+r[0]);
            System.out.println("Second Root = "+r[1]);
        }
    }
}
```

Output:

```
--- exec:3.5.1:exec (default-cli) @ AayushAcharyaJava
Enter value of a: 1
Enter value of b: -4
Enter value of c: 4
First Root = 2.0
Second Root = 2.0
-----
```

Q.28 Create an array of 5 Student objects and print students with age >= 24.

StudentDemo.java

```
import java.util.Scanner;
class Student{
    int age;
    String name;
    static Scanner sc=new Scanner(System.in);
    void input(){
        System.out.print("Enter Name: ");name=sc.nextLine();
        System.out.print("Enter Age: ");age=Integer.parseInt(sc.nextLine());
    }
}
public class StudentDemo{
    public static void main(String[] args){
        Student[] list=new Student[5];
        for(int i=0;i<list.length;i++){System.out.println("Enter details of Student
"+(i+1));list[i]=new Student();list[i].input();}
        System.out.println("Students whose age is greater than or equal to 24:");
    }
}
```

```

        for(Student s:list) if(s.age>=24) System.out.println("Name = "+s.name+", Age = "+s.age);
    }
}

```

Output:

```

j --- exec:3.5.1:exec (default-cli) @ AayushAcharyaJava
Enter details of Student 1
Enter Name: aarjan
Enter Age: 23
Enter details of Student 2
Enter Name: shila
Enter Age: 33
Enter details of Student 3
Enter Name: shiksha
Enter Age: 20
Enter details of Student 4
Enter Name: aayush
Enter Age: 24
Enter details of Student 5
Enter Name: nirajan
Enter Age: 30

Students whose age is greater than or equal to 24:
Name = shila, Age = 33
Name = aayush, Age = 24
Name = nirajan, Age = 30

```

Q.29 Create a Complex class and add real and imaginary parts of two objects.

ComplexDemo.java

```

class Complex{
    int real,img;
    void getSum(Complex c1,Complex c2){
        real=c1.real+c2.real;img=c1.img+c2.img;
        System.out.println("Sum = "+real+" + "+img+"i");
    }
}

public class ComplexDemo{
    public static void main(String[] args){
        Complex c1=new Complex(),c2=new Complex(),c3=new Complex();
        c1.real=17;c1.img=6;c2.real=2;c2.img=10;
        c3.getSum(c1,c2);
    }
}

```

Output:

```

--- exec:3.5.1:exec (default-cli) @ AayushAcharyaJava ---

Sum = 19 + 16i

-----
BUILD SUCCESS

```

Q.30 Create a Time class and return the sum of two Time objects.

TimeDemo.java

```

class Time{
    int hr,min,sec;
    Time Sum(Time t1,Time t2){
        Time t3=new Time();
        t3.sec=t1.sec+t2.sec;
        t3.min=t1.min+t2.min+t3.sec/60;t3.sec%=60;
        t3.hr=t1.hr+t2.hr+t3.min/60;t3.min%=60;
        return t3;
    }
    void display(){System.out.println(hr+":"+min+":"+sec);}
}

```

```

public class TimeDemo{
    public static void main(String[] args){
        Time t1=new Time(),t2=new Time(),t3;
        t1.hr=5;t1.min=40;t1.sec=40;
        t2.hr=4;t2.min=40;t2.sec=50;
        t3=new Time().Sum(t1,t2);
        System.out.print("t1 --> ");t1.display();
        System.out.print("t2 --> ");t2.display();
        System.out.print("Total Sum t3 --> ");t3.display();
    }
}

```

Output:

```

--- exec:3.5.1:exec (default-cli) @ AayushAcharyaJava ---
t1 --> 5:40:40
t2 --> 4:40:50
Total Sum t3 --> 10:21:30
-----
BUILD SUCCESS

```

Q.31 Create an interface Calculate and implement add() and diff().

CalculateDemo.java

```

interface Calculate{
    int add(int x,int y);
    int diff(int x,int y);
}
class Operation implements Calculate{
    public int add(int x,int y){return x+y;}
    public int diff(int x,int y){return x-y;}
}
public class CalculateDemo{
    public static void main(String[] args){
        Operation obj=new Operation();
        System.out.println("Addition = "+obj.add(20,10));
        System.out.println("Subtraction = "+obj.diff(20,10));
    }
}

```

Output:

```

--- exec:3.5.1:exec (default-cli) @ AayushAcharyaJava ---
Addition = 30
Subtraction = 10
-----
BUILD SUCCESS

```

Q.32 Create interface Shape and implement it in Rectangle and Square.

ShapeDemo.java

```

import java.util.Scanner;
interface Shape{
    void get();
    void display();
}
class ShapeInput{
    static Scanner sc=new Scanner(System.in);

```

```

    }
    class Rectangle implements Shape{
        int length,breadth,area;
        public void get(){
            System.out.print("Enter length of Rectangle: ");length=ShapeInput.sc.nextInt();
            System.out.print("Enter breadth of Rectangle: ");breadth=ShapeInput.sc.nextInt();
            area=length*breadth;
        }
        public void display(){System.out.println("Rectangle Area = "+area);}
    }
    class Square implements Shape{
        int side,area;
        public void get(){
            System.out.print("Enter side of Square: ");side=ShapeInput.sc.nextInt();
            area=side*side;
        }
        public void display(){System.out.println("Square Area = "+area);}
    }
    public class ShapeDemo{
        public static void main(String[] args){
            Shape r=new Rectangle(),s=new Square();
            r.get();s.get();r.display();s.display();
        }
    }
}

```

Output:

```

--- exec:3.5.1:exec (default-cli) @ AayushAcharyaJava
Enter Rectangle 1:
Enter length of Rectangle: 10
Enter breadth of Rectangle: 5
Enter Rectangle 2:
Enter length of Rectangle: 20
Enter breadth of Rectangle: 10
Enter Square 1:
Enter side of Square: 12
Enter Square 2:
Enter side of Square: 23

--- Results ---
Rectangle Area = 50
Rectangle Area = 200
Square Area = 144
Square Area = 529

```

Q.33 Create abstract class Fmachine and derive Airplane.

```

AirplaneDemo.java
import java.util.Scanner;
abstract class Fmachine{
    abstract void getdata();
    abstract void putdata();
}
class Airplane extends Fmachine{
    int code,capacity;
    String name;
    Scanner sc=new Scanner(System.in);
    void getdata(){
        System.out.print("Enter Airplane Code: ");code=Integer.parseInt(sc.nextLine());
        System.out.print("Enter Airplane Name: ");name=sc.nextLine();
        System.out.print("Enter Capacity: ");capacity=Integer.parseInt(sc.nextLine());
    }
}

```

```

    void putdata(){
        System.out.println("--- Airplane Details ---");
        System.out.println("Code = "+code);
        System.out.println("Name = "+name);
        System.out.println("Capacity = "+capacity);
    }
}
public class AirplaneDemo{
    public static void main(String[] args){
        Airplane a=new Airplane();
        a.getdata();a.putdata();
    }
}

```

Output:

```

--- exec:3.5.1:exec (default-cli) @ AayushAcharyaJava ---

Enter Airplane Code: Enter Airplane Name: Enter Capacity: --- Airplane Details ---
Code = 17
Name = Aayush Airlines
Capacity = 200
-----
BUILD SUCCESS

```

Q.34 Sort country names and print student names starting with s and ending with a.

CountrySort.java

```

import java.util.Arrays;
public class CountrySort{
    public static void main(String[] args){
        String[] country={"nepal","india","uk","france","australia"};
        Arrays.sort(country);
        System.out.println("Countries in alphabetical order:");
        for(String c:country) System.out.println(c);
    }
}

```

StudentNameFilter.java

```

public class StudentNameFilter{
    public static void main(String[] args){
        String[] names={"Aayush","Swornim","Shiksha","Nilima","Ashim"};
        System.out.println("Students whose name starts with 's' and ends with 'a:");
        for(String n:names){
            String x=n.toLowerCase();
            if(x.startsWith("s")&&x.endsWith("a")) System.out.println(n);
        }
    }
}

```

Output:

```
--- exec:3.5.1:exec (default-cli) @ AayushAcharyaJava ---  
  
Countries in alphabetical order:  
australia  
france  
india  
nepal  
uk  
  
Students whose name starts with 's' and ends with 'a':  
Shiksha  
  
-----  
BUILD SUCCESS
```

Q.35 Write a program for the applet life cycle.

AppletLifeCycle.java

```
import java.applet.Applet;  
import java.awt.Graphics;  
public class AppletLifeCycle extends Applet{  
    String msg="";  
    public void init(){msg="init() called";System.out.println(msg);}  
    public void start(){msg="start() called";System.out.println(msg);}  
    public void paint(Graphics g){msg="paint()  
called";g.drawString(msg,80,100);System.out.println(msg);}  
    public void stop(){System.out.println("stop() called");}  
    public void destroy(){System.out.println("destroy() called");}  
}
```

Output:



Q.36 Create a user input form using AWT.

UserFormAWT.java

```
import java.awt.*;  
import java.awt.event.*;  
public class UserFormAWT extends Frame implements ActionListener{  
    TextField name,age;  
    TextArea address;
```

```

Checkbox male,female,agree;
CheckboxGroup gender;
Choice country;
public UserFormAWT(){
    setTitle("User Input Form");setSize(500,430);setLayout(new FlowLayout());
    name=new TextField("Aayush",20);age=new TextField("20",20);
    gender=new CheckboxGroup();male=new Checkbox("Male",gender,true);female=new
Checkbox("Female",gender,false);
    country=new Choice();country.add("Nepal");country.add("India");country.add("UK");
    address=new TextArea("Kathmandu",4,20);agree=new Checkbox("I agree",true);
    Button b=new Button("Submit");b.addActionListener(this);
    add(new Label("Name:"));add(name);add(new
Label("Age:"));add(age);add(male);add(female);
    add(new Label("Country:"));add(country);add(new
Label("Address:"));add(address);add(agree);add(b);
    setVisible(true);
}
    public void actionPerformed(ActionEvent e){System.out.println("Name:
"+name.getText());}
    public static void main(String[] args){new UserFormAWT();}
}
Output:

```

```

--- exec:3.5.1:exec (default-cli) @ Aayush
---- User Details ----
Name: Aayush
Age: 21
Gender: Male
Country: Nepal
Address: gulmi
Agreed: true

```

Q.37 Create a user input form using Swing.

UserFormSwing.java

```

import javax.swing.*.*;
import java.awt.*.*;
public class UserFormSwing extends JFrame{
    public UserFormSwing(){
        setTitle("User Input Form (Swing)");setSize(500,430);setLayout(new
FlowLayout());setDefaultCloseOperation(EXIT_ON_CLOSE);
        JTextField name=new JTextField("Swornim",20),age=new JTextField("17",20);
        JRadioButton male=new JRadioButton("Male",true),female=new
JRadioButton("Female");
        ButtonGroup group=new ButtonGroup();group.add(male);group.add(female);
        JComboBox<String> country=new JComboBox<>(new
String[]{"Nepal","India","UK"});

```

```

        JTextArea address=new JTextArea("Lalitpur",4,20);
        JCheckBox agree=new JCheckBox("I agree",true);
        add(new JLabel("Name:"));add(name);add(new
        JLabel("Age:"));add(age);add(male);add(female);
        add(new JLabel("Country:"));add(country);add(new
        JLabel("Address:"));add(address);add(agree);add(new JButton("Submit"));
        setVisible(true);
    }
    public static void main(String[] args){new UserFormSwing();}
}

```

Output:

Q.38 Create a user input form using JavaFX.

UserFormJavaFX.java

```

import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.layout.FlowPane;
import javafx.stage.Stage;
public class UserFormJavaFX extends Application{
    public void start(Stage s){
        TextField name=new TextField("Shiksha"),age=new TextField("20");
        RadioButton male=new RadioButton("Male"),female=new RadioButton("Female");
        ToggleGroup group=new
        ToggleGroup();male.setToggleGroup(group);female.setToggleGroup(group);female.setSelected(true);
        ComboBox<String> country=new
        ComboBox<>();country.getItems().addAll("Nepal","India","UK");country.setValue("Nepal")
        ;
        TextArea address=new TextArea("Bhaktapur");
        FlowPane root=new FlowPane(10,10,new Label("Name:"),name,new
        Label("Age:"),age,male,female,country,address,new CheckBox("I agree"),new
        Button("Submit"));
        s.setTitle("JavaFX User Form");s.setScene(new Scene(root,420,300));s.show();
    }
    public static void main(String[] args){launch(args);}
}

```

Output:

Name: Age: ☐ Male ☒ Female

☐ I agree

Q.39 Calculate simple interest on button click using TextField values.

SimpleInterest.java

```
import java.awt.*;
import java.awt.event.*;
public class SimpleInterest extends Frame implements ActionListener{
    TextField p,r,t,si;
    public SimpleInterest(){
        setTitle("Simple Interest Calculator");setSize(520,220);setLayout(new FlowLayout());
        p=new TextField("1000",10);r=new TextField("10",10);t=new TextField("3",10);si=new
        TextField(10);
        Button b=new Button("Calculate");b.addActionListener(this);
        add(new Label("Principal:"));add(p);add(new Label("Rate:"));add(r);add(new
        Label("Time:"));add(t);add(b);add(new Label("Simple Interest:"));add(si);
        setVisible(true);
    }
    public void actionPerformed(ActionEvent
    e){si.setText(String.valueOf(Double.parseDouble(p.getText())*Double.parseDouble(r.getTex
    t())*Double.parseDouble(t.getText())/100));}
    public static void main(String[] args){new SimpleInterest();}
}
```

Output:

Simple Interest Calculator

Principal: Rate:

Time:

Simple Interest:

Q.40 Move an image left, right, top and bottom using keyboard keys.

MoveImage.java

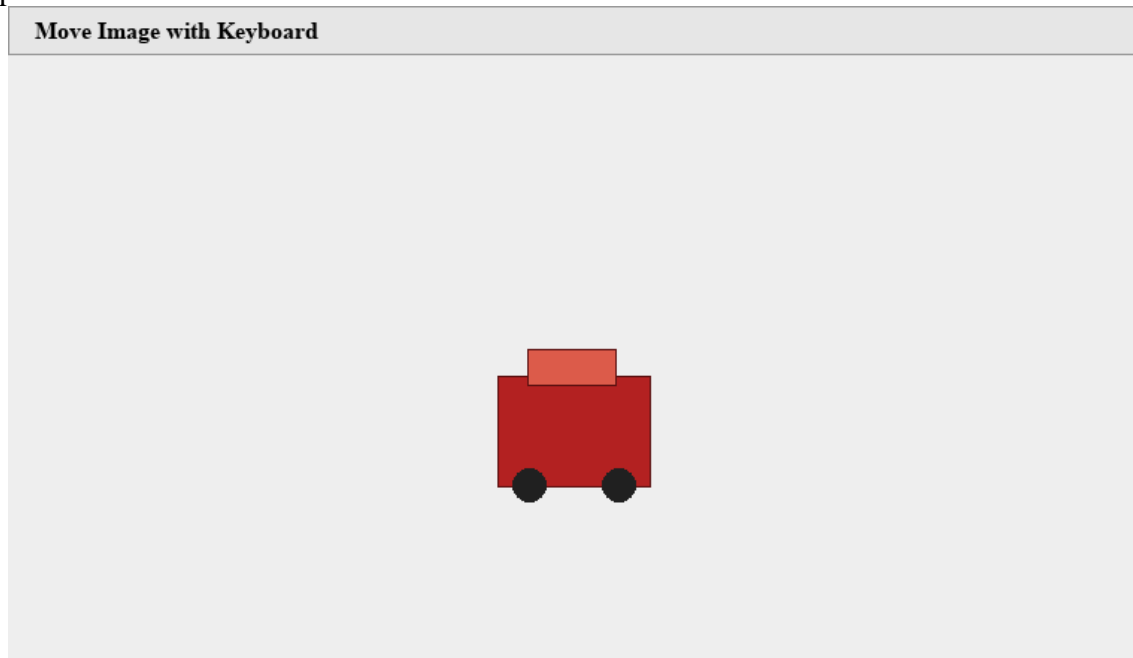
```
import java.awt.*;
import java.awt.event.*;
public class MoveImage extends Frame implements KeyListener{
    int x=220,y=220;
    Image img;
    public MoveImage(){
        setTitle("Move Image with Keyboard");setSize(500,500);setVisible(true);
```

```

        img=Toolkit.getDefaultToolkit().getImage("car.png");
        addKeyListener(this);
    }
    public void paint(Graphics g){g.drawImage(img,x,y,100,80,this);}
    public void keyPressed(KeyEvent e){
        if(e.getKeyCode()==KeyEvent.VK_LEFT)x-=10;
        if(e.getKeyCode()==KeyEvent.VK_RIGHT)x+=10;
        if(e.getKeyCode()==KeyEvent.VK_UP)y-=10;
        if(e.getKeyCode()==KeyEvent.VK_DOWN)y+=10;
        repaint();
    }
    public void keyReleased(KeyEvent e){}
    public void keyTyped(KeyEvent e){}
    public static void main(String[] args){new MoveImage();}
}

```

Output:



Q.41 Insert record into database on button click with validation.

InsertStudent.java

```

import javax.swing.*.*;
import java.awt.*.*;
import java.awt.event.*;
import java.sql.*;

public class InsertStudent extends JFrame implements ActionListener{
    JTextField roll,first,last;
    public InsertStudent(){
        setTitle("Student Record Insert");setSize(500,170);setLayout(new
        FlowLayout());setDefaultCloseOperation(EXIT_ON_CLOSE);
        roll=new JTextField("17",8);first=new JTextField("Aayush",12);last=new
        JTextField("Acharya",12);
        JButton insert=new JButton("Insert");insert.addActionListener(this);
        add(new JLabel("Roll:"));add(roll);add(new JLabel("First Name:"));add(first);add(new
        JLabel("Last Name:"));add(last);add(insert);
    }
}

```

```

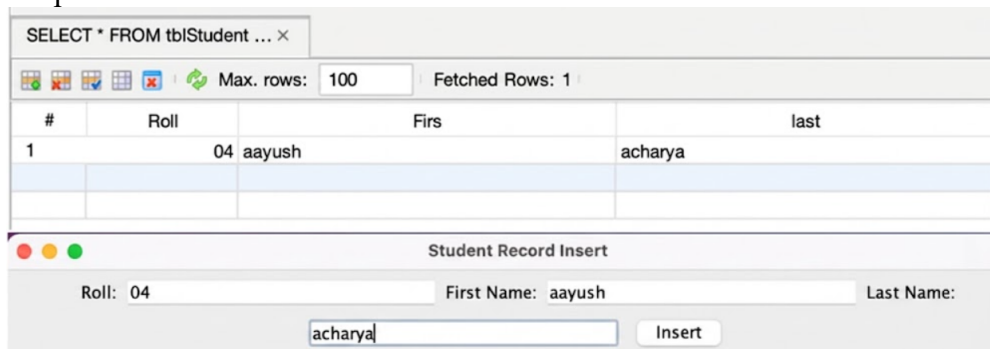
        setVisible(true);
    }
    public void actionPerformed(ActionEvent e){

if(roll.getText().isEmpty()||first.getText().isEmpty()||last.getText().isEmpty()){JOptionPane.s
howMessageDialog(this,"All fields are required");return;}
        try(Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/Test","root","")){
            PreparedStatement ps=con.prepareStatement("insert into
tblStudent(Roll,FirstName,LastName) values(?,?,?)");

ps.setInt(1,Integer.parseInt(roll.getText()));ps.setString(2,first.getText());ps.setString(3,last.g
etText());ps.executeUpdate();
            JOptionPane.showMessageDialog(this,"Record Inserted");
        }catch(Exception ex){JOptionPane.showMessageDialog(this,ex.getMessage());}
    }
    public static void main(String[] args){new InsertStudent();}
}

```

Output:



Q.42 Fetch records from database and display them in JTable.

ViewStudent.java

```

import javax.swing.*.*;
import javax.swing.table.DefaultTableModel;
import java.sql.*;

public class ViewStudent extends JFrame{
    DefaultTableModel model=new DefaultTableModel(new String[]{"Roll","First
Name","Last Name"},0);
    public ViewStudent(){
        setTitle("Student Records Table");setSize(520,300);add(new JScrollPane(new
JTable(model)));setDefaultCloseOperation(EXIT_ON_CLOSE);
        fetch();setVisible(true);
    }
    void fetch(){
        try(Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/Test","root","");
            Statement st=con.createStatement();ResultSet rs=st.executeQuery("select
Roll,FirstName,LastName from tblStudent")){
            while(rs.next()) model.addRow(new
Object[]{rs.getInt(1),rs.getString(2),rs.getString(3)});

```

```

    } catch (Exception e) { JOptionPane.showMessageDialog(this, e.getMessage()); }
    }
    public static void main (String[] args) { new ViewStudent(); }
}

```

Output:



Roll No	First Name	Last Name
47	swornima	singh
37	shiksha	karki
32	saranjana	marasini
20	nilima	shrestha
04	aayush	acharya

Q.43 Servlet program to find prime numbers between a and b.

index.html

```

<!DOCTYPE html>
<html>
<head><title>Prime Finder</title></head>
<body>
<h2>Find Prime Numbers in a Range</h2>
<form action="PrimeServlet" method="get">
Start (a): <input type="number" name="a"><br><br>
End (b): <input type="number" name="b"><br><br>
<input type="submit" value="Find Primes">
</form>
</body>
</html>

```

PrimeServlet.java

```

import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;
@WebServlet("/PrimeServlet")
public class PrimeServlet extends HttpServlet {
    protected void doGet (HttpServletRequest req, HttpServletResponse res) throws
ServletException, IOException {
        int a=Integer.parseInt(req.getParameter("a"));
        int b=Integer.parseInt(req.getParameter("b"));
        res.setContentType("text/html");
        PrintWriter out=res.getWriter();
        out.println("<h3>Prime numbers between "+a+" and "+b+" are:</h3>");
        for(int i=a;i<=b;i++) if(isPrime(i)) out.print(i+" ");
    }
    boolean isPrime(int n){
        if(n<2)return false;
        for(int i=2;i<=Math.sqrt(n);i++) if(n%i==0)return false;
        return true;
    }
}

```

Output:



Prime numbers between 12 and 20 are:

13 17 19

[Back](#)

Q.44 Servlet program to calculate simple interest and display session result in next servlet.
index.html

```
<!DOCTYPE html>
<html>
<head><title>Simple Interest</title></head>
<body>
<h2>Simple Interest Calculator</h2>
<form action="CalculateServlet" method="post">
Principal: <input type="number" name="p"><br><br>
Rate: <input type="number" name="r"><br><br>
Time: <input type="number" name="t"><br><br>
<input type="submit" value="Calculate and Save">
</form>
</body>
</html>
```

CalculateServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;
@WebServlet("/CalculateServlet")
public class CalculateServlet extends HttpServlet {
    protected void doPost(HttpServletRequest req, HttpServletResponse res) throws
ServletException, IOException {
        double p=Double.parseDouble(req.getParameter("p"));
        double r=Double.parseDouble(req.getParameter("r"));
        double t=Double.parseDouble(req.getParameter("t"));
        req.getSession().setAttribute("si",p*r*t/100);
        res.sendRedirect("DisplayServlet");
    }
}
```

DisplayServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;
@WebServlet("/DisplayServlet")
public class DisplayServlet extends HttpServlet {
    protected void doGet(HttpServletRequest req, HttpServletResponse res) throws
ServletException, IOException {
        res.setContentType("text/html");
        PrintWriter out=res.getWriter();
        out.println("<h2>The Simple Interest stored in Session is:
```

```

"+req.getSession().getAttribute("si")+"</h2>");
    out.println("<a href='index.html'>Calculate Again</a>");
}
}

```

Output:



Q.45 JSP program to print your name 10 times.

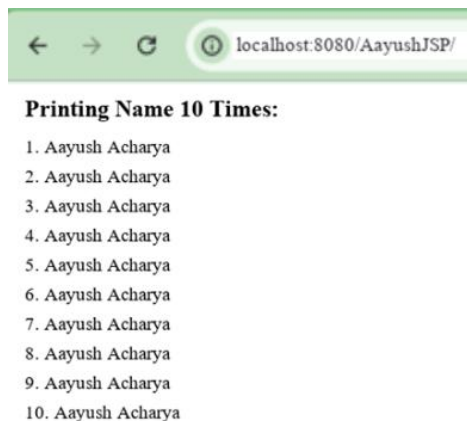
index.jsp

```

<%@ page contentType="text/html; charset=UTF-8" %>
<!DOCTYPE html>
<html>
<head><title>Name Repeater</title></head>
<body>
<h2>Printing Name 10 Times:</h2>
<ul>
<% for(int i=1;i<=10;i++){ %>
<li><%= i %>. Aayush Acharya</li>
<% } %>
</ul>
</body>
</html>

```

Output:



Q.46 JSP program to insert record into database from HTML parameters.

index.jsp

```

<!DOCTYPE html>
<html>
<head><title>Register Student</title></head>
<body>
<h2>Register Student</h2>
<form action="insert.jsp" method="post">
Username: <input type="text" name="uName"><br><br>
Password: <input type="password" name="pWord"><br><br>
<input type="submit" value="Register">

```

```

</form>
</body>
</html>

```

insert.jsp

```

<%@ page import="java.sql.*" %>
<%
String user=request.getParameter("uName");
String pass=request.getParameter("pWord");
try {
    Class.forName("com.mysql.cj.jdbc.Driver");
    Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/Test","root","");
    PreparedStatement ps=con.prepareStatement("insert into tblStudent(UserName,Password)
values(?,?)");
    ps.setString(1,user);ps.setString(2,pass);
    int result=ps.executeUpdate();
    out.println(result>0?"Record inserted successfully for: "+user:"Failed to insert record.");
    ps.close();con.close();
} catch(Exception e){out.println(e.getMessage());}
%>

```

Output:



Q.47 Login form with database validation and session redirect.

login.jsp

```

<!DOCTYPE html>
<html>
<head><title>Login Page</title></head>
<body>
<h2>Login</h2>
<form action="loginprocess.jsp" method="post">
Username: <input type="text" name="uName"><br><br>
Password: <input type="password" name="pWord"><br><br>
<input type="submit" value="Login">
</form>
</body>
</html>

```

loginprocess.jsp

```

<%@ page import="java.sql.*" %>
<%
String user=request.getParameter("uName");
String pass=request.getParameter("pWord");
try {
    Class.forName("com.mysql.cj.jdbc.Driver");
    Connection

```

```

con=DriverManager.getConnection("jdbc:mysql://localhost:3306/Test","root","");
PreparedStatement ps=con.prepareStatement("select * from tblStudent where UserName=?
and Password=?");
ps.setString(1,user);ps.setString(2,pass);
ResultSet rs=ps.executeQuery();

```

```

if(rs.next()){session.setAttribute("UserName",user);response.sendRedirect("welcome.jsp");}
else out.println("Invalid User Name Or Password");
rs.close();ps.close();con.close();
}catch(Exception e){out.println(e.getMessage());}
%>

```

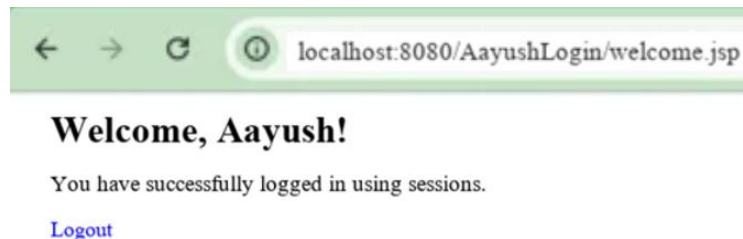
welcome.jsp

```

<%@ page contentType="text/html; charset=UTF-8" %>
<%
String user=(String)session.getAttribute("UserName");
if(user==null){response.sendRedirect("login.jsp");return;}
%>
<!DOCTYPE html>
<html>
<head><title>Welcome</title></head>
<body>
<h1>Welcome, <%= user %>!/h1>
<p>You have successfully logged in using sessions.</p>
<a href="logout.jsp">Logout</a>
</body>
</html>

```

Output:



Q.48 In welcome.jsp, display a special message if the session username is john.

welcome.jsp

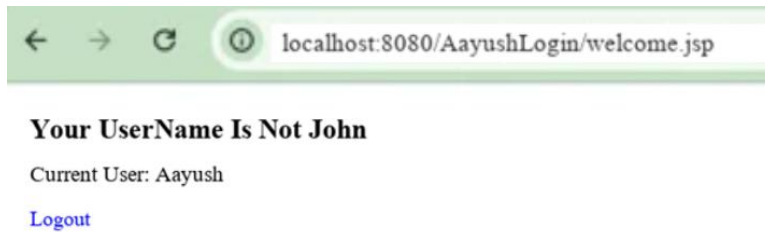
```

<%@ page contentType="text/html; charset=UTF-8" %>
<%
String user=(String)session.getAttribute("UserName");
if(user==null){response.sendRedirect("login.jsp");return;}
%>
<!DOCTYPE html>
<html>
<head><title>Welcome Page</title></head>
<body>
<%
if(user.equalsIgnoreCase("john")) out.println("<h2>Your Are John</h2>");
else out.println("<h2>Your UserName Is Not John</h2><p>Current User: "+user+"</p>");
%>

```

```
<a href="logout.jsp">Logout</a>
</body>
</html>
```

Output:



Q.49 Show output when the session value is john.

welcome.jsp

```
<%@ page contentType="text/html; charset=UTF-8" %>
<%
String user=(String)session.getAttribute("UserName");
if(user==null){response.sendRedirect("login.jsp");return;}
%>
<!DOCTYPE html>
<html>
<head><title>Welcome Page</title></head>
<body>
<%
if(user.equalsIgnoreCase("john")) out.println("<h2>Your Are John</h2>");
else out.println("<h2>Your UserName Is Not John</h2><p>Current User: "+user+"</p>");
%>
<a href="logout.jsp">Logout</a>
</body>
</html>
```

Output:

